Project rules

Groups

max 4

students per group

- Register members and project in the **next practical class**
- At most **two groups** per project
- Groups that share a project study a different cohort, target or tissue

Preparation

1

biological question drives each project

- Start from the **question**, not from a tool
- Use **open-access data** only
- **Justify every method** you choose
- Run the **dataset check** before Checkpoint 1

Assessment

15 Dec

final evaluation: report and defence

- **Checkpoint 1:** week of 23 Nov
- **Checkpoint 2:** 4 Dec
- Both checkpoints **count toward the grade**

Research here is question-driven, not tool-driven

Every choice in the project, from the dataset to the method, traces back to one biological question.

1

The problem

State the biological or biomedical problem explicitly, and the role bioinformatics plays in solving it.

2

The data

Identify the dataset and data sources you will use, with names, accession numbers or URLs.

3

The analysis

Describe the prediction or comparison you will make, and justify the tools you selected.

Example: “Which processes separate HER2-negative breast tumours that respond to chemotherapy from those that do not?” The question picks the data (GEO GSE192341) and the method (DESeq2).

Three weeks to a final evaluation on 15 December

Weeks 1 and 2 each end with a graded checkpoint in the practical class.

1

WEEK 1

Question and data

Write the research question, access the data and run the dataset check.

Checkpoint 1
week of 16 Nov

2

WEEK 2

Run the analysis

Compare at least two methods; run the baseline and the main analysis on real data.

Checkpoint 2
week of 30 Nov

3

WEEK 3

Interpret and report

Final results and limitations; report, tables and figures; prepare the defence.

4

15 DEC

Final evaluation

Submit the report and present the defence.

Checkpoints: scores the progress regarding milestone but counts toward the grade (10% each)

Minimum outcomes at each checkpoint and the final

Bring these to the practical class: they are the minimum each group must show, not the target.

CHECKPOINT 1 · WEEK OF

16 Nov

Question and data in place

- **Research question** in one sentence
- **Dataset check** runs on the real data
- **Baseline method** and its first number
- **Roles split** and a plan B for the data

CHECKPOINT 2

30 Nov

Analysis running

- **Main analysis** run end to end
- At least **two methods** compared
- **First result** set against the baseline
- **Limitations** identified

FINAL EVALUATION

15 Dec

Report and defence

- Report **answers the question**
- **Tables and figures** support each claim
- **Limitations** and next steps stated
- **Defence** of methods and results

Checkpoints score four criteria from 0 to 2

The instructor scores every group at both checkpoints, and both scores count toward the grade.

Criterion	Level 0	Level 1	Level 2
Progress	No progress shown	Limited progress	Clear, documented progress
Technical quality	Missing or unsuitable output	Problems partly solved	Technically sound analysis
Interpretation	No interpretation	Superficial interpretation	Well-argued interpretation
Group organisation	Tasks undefined	Unclear division of work	Clear roles and responsibilities

Projects List

#	Project
1	Network position or publication count?
2	Enzymatic degradation of polyesters
3	Thermostability of industrial enzymes
4	Which substrate does this enzyme accept?
5	From a GWAS locus to the causal gene
6	Are single-cell foundation models worth it?
7	Repurposing approved drugs against one target
8	Do the ADMET leaderboard gains survive?
9	Generative design versus the catalogue
10	Natural products as a screening library

Network position or publication count?

1

Research question: Does a gene's position in the protein interaction network predict disease association, or does it only predict how much the gene has been studied?

Dataset

- STRING v12, human protein–protein interactions with per-channel confidence scores
- Open Targets Platform: gene–disease associations, used as the labels
- NCBI gene2pubmed: publications per gene, the confounder to control for
- Reactome and Gene Ontology for pathway annotation of the communities found

Context

Hub genes in protein interaction networks are routinely put forward as drug targets, and centrality is used to rank them. Those networks are assembled from published experiments, so the genes that have been studied most carry the most edges — and the disease labels come from the same literature.

The problem: Separate biology from study bias: control for publication count, and report whether network position still predicts disease association once it is held constant.

STRING: Szklarczyk et al., Nucleic Acids Res 2023 · Open Targets Platform · NCBI gene2pubmed

Enzymatic degradation of polyesters

Research question: How well do sequence-based predictors of enzymatic polyester degradation generalise to unseen enzyme families and new polymer chemistries?

Dataset

- PAZy: biochemically verified plastic-active enzymes (477 unique entries), PET and PUR
- PlasticDB and PMBD for further reported enzymes and organisms; keep negative results
- UniProt for sequences; MMseqs2 to cluster enzymes into families for the held-out split
- Polymer side: repeat-unit descriptors from RDKit, optionally polyBERT embeddings

Context

Enzymatic recycling of PET already runs at demonstration scale, so companies are searching sequence databases for enzymes that attack other polyesters and polyurethanes. Only a few hundred plastic-active enzymes have been verified in the laboratory, and they cluster in a handful of families.

The problem: Measure how far a sequence-based predictor travels: hold out whole enzyme families and polymer chemistries it never saw, and report what accuracy is left.

PAZy: Buchholz et al., Proteins 90:1443 (2022), pazy.eu · polyBERT: Kuenneth & Ramprasad, Nat Commun 2023

Thermostability of industrial enzymes

3

Research question: Can a protein language model predict an enzyme's melting temperature in families it never saw — and does it beat amino-acid composition?

Dataset

- FLIP meltome splits (mixed, human, human-cell): sequence and melting temperature, ready-split
- Meltome Atlas (Jarzab et al. 2020) for the underlying proteome-wide measurements
- FireProtDB or ProThermDB if the team prefers stability changes on mutation instead
- ESM-2 embeddings from Hugging Face; UniProt for taxonomy and family annotation

Context

An industrial enzyme is useless above its melting temperature, so Tm decides whether a catalyst survives a process run at 60 or 80 degrees. Measuring it is slow, which is why protein language models are used to predict it — usually benchmarked on random splits that put close relatives on both sides.

The problem: Hold out whole families and answer two things: does the language model still predict Tm, and does it beat plain amino-acid composition by enough to justify its cost?

FLIP: Dallago et al., NeurIPS Datasets & Benchmarks 2021 · Meltome Atlas: Jarzab et al., Nat Methods 17:495 (2020)

Which substrate does this enzyme accept?

Research question: Does an enzyme–substrate prediction model still work for enzyme families and substrate chemistries it never saw?

Dataset

- ESP dataset (Kroll et al. 2023): enzyme–small molecule pairs with sampled negatives
- BRENDA and Rhea for additional curated pairs; ChEBI for substrate structures
- ESM-2 for the enzyme side, ECFP4 or a graph network for the molecule side
- GO annotations for the family labels used to build the held-out split

Context

Sequencing assigns most enzymes a family by similarity and stops there: the substrate stays unknown, which blocks both metabolic reconstruction and enzyme engineering. Models such as ESP predict whether an enzyme accepts a given small molecule, trained on curated pairs with computationally sampled negatives.

The problem: Test the model on enzyme families and substrate chemistries absent from training, and show how much of its reported performance comes from the way negatives were sampled.

From a GWAS locus to the causal gene

Research question: In a GWAS locus, is the nearest gene the causal gene — and when it is not, what evidence finds the right one?

Dataset

- GWAS Catalog associations and, where available, fine-mapped credible sets
- Open Targets Genetics locus-to-gene gold standards: the labels, with their own biases
- GTEx v8 eQTL for the expression evidence; Ensembl for gene coordinates and consequences
- Open Targets Platform for druggability of the genes that come out on top

Context

A GWAS hit marks a region of the genome, not a gene. Papers default to the gene nearest the lead variant, and that shortcut has been wrong in well-documented cases: the obesity signal inside FTO acts on IRX3 and IRX5. Which gene is picked decides which target a programme pursues.

The problem: Across loci where the causal gene is known, measure how often the nearest gene is right, and establish which evidence — eQTL, chromatin contact, distance — finds the gene when it is wrong.

Open Targets Genetics: Mountjoy et al., Nat Genet 53:1527 (2021) · GWAS Catalog · GTEx Consortium, Science 2020

Are single-cell foundation models worth it?

Research question: Do foundation-model embeddings annotate cell types in a study the model never saw better than a well-tuned classical pipeline?

Dataset

- CELLxGENE Census: one tissue, several independent studies, with donor and assay metadata
- Tabula Sapiens as a second source of held-out studies
- CellTypist and scVI as the classical comparators
- scGPT or Geneformer weights from Hugging Face for the zero-shot embeddings

Context

Cell type annotation is the first step of almost every single-cell study, and it still rests on marker genes and manual curation. Foundation models trained on tens of millions of cells offer to do it without labels, but published comparisons often tune the new model and leave the classical pipeline at its defaults.

The problem: Run both on studies neither has seen, give the classical pipeline the same care as the foundation model, and state whether the extra compute buys accuracy.

CZ CELLxGENE Discover: Program et al., Nucleic Acids Res 2025 · CellTypist: Domínguez Conde et al., Science 2022

Repurposing approved drugs against one target

7

Research question: For one target of your choosing, which approved drugs does a ligand-based model rank as binders — and does the ranking recover the drugs already approved against that target?

Dataset

- ChEMBL bioactivity for your target, curated to pIC50: actives and inactives for training
- DrugBank approved drugs with structures: the library to rank
- Drugs already approved against that target: the positive control
- UniProt and PDB entries for the binding site

Context

An approved drug has cleared safety and manufacturing, so a new indication reaches patients years sooner than a new molecule.

The problem: Fix one target, rank every approved drug against it, and test the ranking: if the drugs already approved against that target do not surface near the top, it is not evidence.

Pick one target

- A2A receptor (ADORA2A)
- TRK family (NTRK1–3)
- ABL1 or BRAF
- PARP1
- D2 dopamine receptor or SLC6A4
- Acetylcholinesterase
- DHFR

ChEMBL: Zdrazil et al., Nucleic Acids Res 2024 · DrugBank: Knox et al., Nucleic Acids Res 2024 · Open Targets Platform

Do the ADMET leaderboard gains survive?

Research question: Do the reported gains on ADMET benchmarks survive a harder split and an honest error bar — or are the leaders inside the noise?

Dataset

- Therapeutics Data Commons ADMET group: 22 datasets, official splits, five seeded runs
- Endpoint sizes and assay provenance, recorded per dataset — many have a few hundred molecules
- RDKit descriptors and ECFP4 for the baseline; a GNN and ChemBERTa as the challengers
- ChEMBL for extra assay context when an endpoint looks suspiciously heterogeneous

Context

Absorption, toxicity and clearance kill most drug candidates, so ADMET predictions decide which molecules advance and which are dropped. The Therapeutics Data Commons leaderboards are the scoreboard the field quotes, yet many of its endpoints hold only a few hundred molecules and the gaps between leading entries are small.

The problem: Re-run the leaders and a plain descriptor baseline over repeated seeds and a scaffold split, with error bars, and report which published gains survive.

Therapeutics Data Commons: Huang et al., NeurIPS Datasets & Benchmarks 2021 · MoleculeNet: Wu et al., Chem Sci 2018

Generative design versus the catalogue

9

Research question: Does a generative model propose better molecules than simply screening a catalogue with the same scoring function?

Dataset

- Target: one from the project 7 list that has a co-crystal structure — CDK2, JAK2 or A2A
- ChEMBL bioactivity for that target, curated to pIC50 — the training set for both QSAR models
- ZINC20 subset as the catalogue to screen: the baseline that is cheap and often wins
- PDB structure with a co-crystallised ligand, for the docking confirmation
- RDKit for SA score, QED and novelty against the training set

Context

Generative models design molecules against a scoring function that is itself a model trained on the same bioactivity data, and they can exploit it: high predicted scores, no real affinity. Buying from a catalogue of purchasable compounds scored the same way is the cheap alternative, and it is rarely reported as the comparison.

The problem: Give generation and catalogue screening the same scoring function and the same budget, judge the output by docking, synthetic accessibility and novelty, and say whether designing molecules earns its place.

ChEMBL: Zdrazil et al., Nucleic Acids Res 2024 · model exploitation: Renz et al., Drug Discov Today Technol 2019 · MOSES: Polykovskiy et al., Front Pharmacol 2020

Natural products as a screening library

Research question: Does a natural-product library enrich for actives against one target better than a synthetic library of the same size?

Dataset

- Target: one from the project 7 list that DUD-E covers — acetylcholinesterase, PARP1 or A2A
- COCONUT 2.0: open collection of natural products, with structures and provenance
- LOTUS and NPASS for organism of origin and reported activities
- ChEMBL actives plus DUD-E decoys for the enrichment measurement
- PDB target with a co-crystallised ligand; ZINC20 as the synthetic comparator

Context

Natural products occupy a different region of chemical space from synthetic screening decks: heavier, richer in sp3 centres and stereochemistry, and the origin of a large share of approved small-molecule drugs. Whether that distinctiveness helps in a virtual screen against any particular target is untested.

The problem: Screen a natural-product library and a synthetic library of the same size against one target with the same method, and compare how well each enriches known actives over decoys.

COCONUT 2.0: Chandrasekhar et al., J Cheminform 2025 · LOTUS: Rutz et al., eLife 2022 · DUD-E: Mysinger et al., J Med Chem 2012